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Analysing the 'dark side' of Nrf2 activation using RNA sequencing

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Abstract

Introduction

NRF2 is a transcription factor regulated by Keap1. A “master regulator” of cytoprotective genes and antioxidant response, it is considered a promising therapeutic avenue for chronic diseases. Using RNA-Sequencing, investigation of the ‘dark side’ of NRF2 activation was performed to provide insight into the safety and efficacy of its use as a treatment.

Methodology

RNA-sequencing reads from mouse hepatocytes were analysed to identify differentially expressed genes (DEGs) between control and CRISPR-Cas9 edited Keap1 mutant conditions. Data quality assessment was performed using FastQC before mapping to the mm9 genome via RNA STAR. After validation of mapping using Read Distribution and featureCounts, DEG and GO analysis was performed using DESeq2 and GProfiler.

Results

Data quality was validated as suitable for analysis. A 3’ bias and high duplication level was identified prior to and post mapping at 64.4-77.7% of reads. DEG analysis identified 410 genes (Padj-value <0.01, log fold change >2). Significant relevant GO terms were identified, including glutathione transferase activity (Padj value 4.47×10^{-7}) and cell proliferation (Padj value 3.177×10^{-5}) with related genes GSTA1, GSTP1/2 and BMPR1a up to 5.53-fold upregulated.

Conclusions

The project confirmed some ‘dark sides’ of NRF2 activation. Glutathione activity and associated gene modulation support NRF2 driving chemoresistance and neoplastic disease. This was further substantiated by GO analysis identifying cell proliferation and differentiation terms and associated BMPRa1 gene upregulation. Data issues such as 3’ bias and high duplication levels indicate a bias towards short reads and possible rRNA contamination that may have influenced this projects analysis.

Introduction

Nuclear factor erythroid 2-related factor 2 (NRF2) is a bZIP transcription factor that forms part of the “Cap’N’Collar’ family that moderate adaptive responses to cellular stress (Sykiotis & Bohmann, 2010). In homeostatic conditions, NRF2 is tightly regulated and sequestered from the nucleus by formation of a complex with Kelch-like ECH-associated protein 1 (Keap1) which targets NRF2 for ubiquitination by a CUL3 ligase, keeping its basal level low in most cells (Ma, 2013). Under cellular exposure to oxidative stress, Keap1’s highly sensitive cysteine residues are oxidised leading to dissociation from NRF2 (Zhang & Hannink, 2003). This allows NRF2 to escape ubiquitination and subsequent proteasomal degradation, accumulate, and translocate to the nucleus where it forms heterodimers with sMAF and binds to the Antioxidant Response Element (ARE) (Mathew & Hammarsten, 2016). The ARE is a cis-acting DNA regulatory element situated in the promoter region of NRF2 target genes (Raghunath et al.,

2018). Functions of these target genes are varied, with NRF2 considered a “master regulator” of cytoprotective genes and response to oxidative stress (Tonelli et al., 2018) including genes such as glutathione s-transferases (GSTP) that metabolise genotoxic foreign complexes (Cui et al., 2020) and NAD(P)H dehydrogenase 1 (NQO1) that acts as a superoxide reductase (Ross & Siegel, 2017).

NRF2 has garnered significant attention in recent years for its multifaceted role in protecting against various chronic conditions and potentially extending lifespan. Notably, it has been shown to treat multiple sclerosis using dimethyl fumarate (DMF) as an activator (Carlström et al., 2019), combat Parkinson’s caused by ROS production (Yang et al., 2022), regulate lifespan in insect models (Sykiotis et al.,2011), and be protective against cardiovascular disease in obese patients (Gutiérrez-Cuevas et al., 2022). While NRF2 activation has shown great potential for therapeutic applications, recent studies have raised concerns about potential negative effects and ‘dark sides’ it may have, including suggestions that NRF2 may be an oncogene (Rojo de la Vega et al., 2018), promote proliferation in neoplastic liver disease (Orrù et al., 2020), mediate myocardial damage and dysfunction (Zang et al.,2020), and enhance resistance to chemotherapeutic drugs (Wang et al., 2008).

To investigate this ‘dark side’ of NRF2 activation, RNA-sequencing reads from mouse hepatocytes edited for inactivating mutations in Keap1 have been analysed. Keap1 knockout studies in mice have validated that this method will result in constitutive activation of NRF2 (Wakabayashi et al.,2003). This approach allowed this project to examine the safety implications and any potential adverse effects of proposed therapeutics using activation of NRF2 by identifying any differentially expressed genes under these conditions.

Methods

RNA-Sequencing

This project entailed the usage of previously generated RNA-Sequencing data obtained from mouse primary hepatocytes carrying inactivating mutations within the Keap1 gene. CRISPR/Cas9 genome editing was used to induce the mutations before the cells were cultured for 5 days prior to harvesting and total RNA extraction. Three biological replicates were sequenced from control and Keap1 mutant cells producing single end 50bp reads.

Data Analysis and Quality Control

Data analysis was carried out using the open-source Galaxy platform and software packages (Galaxy Community, 2022). The RNA-sequencing data was initially assessed for quality using FastQC (Andrews, 2010). Cutadapt was used to identify and remove known adapter sequences and trim low-quality bases at a quality threshold of 20 (Martin, 2011). The RNA-sequencing reads were then aligned with the Mouse Genome Sequencing Consortium *mus musculus* 9 assembly release 37 using RNA Star (Dobin, 2014). IGV visualisation of RNA Star BAM files was used to validate mapping of reads to exons. Quality of mapping was further assessed using a pipeline of Galaxy packages: MarkDuplicates, Samtools idxstats, Gene Body Coverage, and Read Distribution (Broad Institute, 2017)

An estimation of the strandness of the RNA-sequencing library was achieved using Infer Experiment on 200,000 reads sampled from the BAM files (Wang et al., 2012). featureCounts was used to count the number of reads per annotated gene (Liao et al., 2013).

Differential Gene Expression Analysis

Raw gene counts were normalised for sequencing depth and library composition via DESeq2 (Love et al., 2014). To determine differentially expressed genes via DESeq2 functions, the Wald test was applied to determine if changes were statistically significant (Wolfgang Huber, 2017). The Benjamini Hochberg correction was then applied to control the expected False Discovery Rate and assign Padj-values to each gene. The list of genes was filtered using a significance threshold of adjusted P-value <0.01 and a log fold change >2 before annotation with their names and pathway analysis. The Volcano Plot package was used to visualise the differentially expressed genes (Doyle, 2022).

Gene ontology enrichment analysis was performed using GProfiler to categorise the differentially expressed genes by function (Raudvere et al., 2019). GProfiler uses Fisher's one tailed test to determine an enrichment P-value measuring the randomness of inputted genes intersecting with the ontology terms selected. GProfiler functions were also used to assess enrichment for transcription factor binding sites.

Results

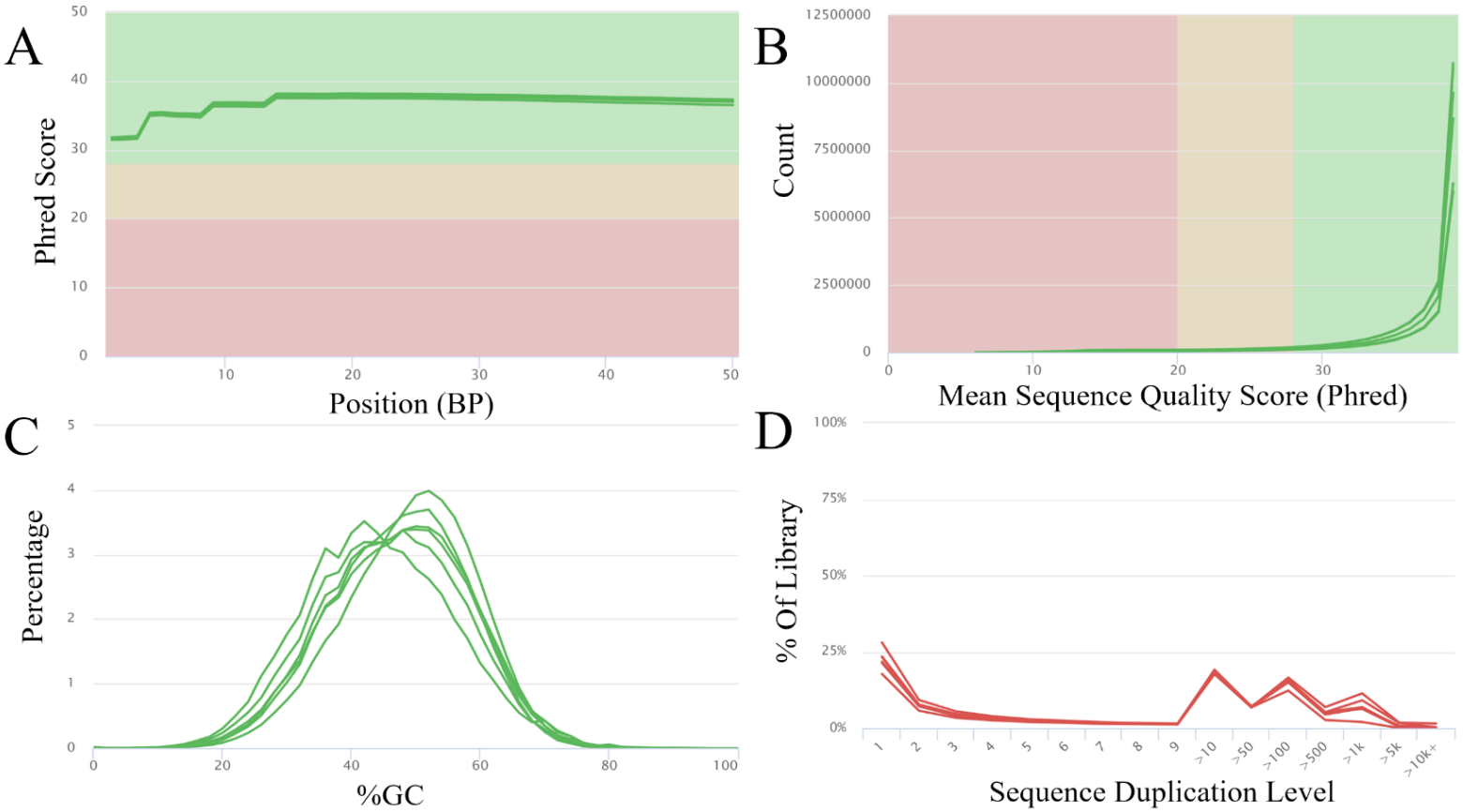

Figure 1. **Assessment of RNA-seq read quality through FastQC quality control analysis**. Summary plots assembled from FastQC analysis on both control and experimental data sets using MultiQC. Mean read quality per base position (A) and per sequence (B) are displayed for both mutant and control conditions. Individual dataset Phred scores are plotted as green lines, with green zones representing very good quality calls, orange representing calls of reasonable quality, and red representing poor quality or failure. Distribution of guanine cystine (GC) base content across sequences are plotted as % per data set as green lines (C). Sequence duplication levels per condition plotted as red lines displayed as a % of the library (D).

Initially, quality control of the RNA-Sequencing reads produced from control and Keap1 mutant hepatocytes was undertaken to determine the suitability of the raw data for further analysis and to identify any errors that may have occured during sequencing.

FastQC analysis validated that the reads were above the good quality threshold. Per base sequence Phred scores were consistently high, ranging from 31.45-38.15 across all datasets (Figure 1A). Phred scores are used to indicate the measure of base quality in RNA-Sequencing, with scores >30 indicating a base call accuracy of >99.9%. Quality per sequence was also assessed, with the majority of sequences having a “good quality” mean Phred score of >20 (Figure 1B).

Guanine cystine (GC) base content was found to be consistent across all control samples, averaging 50-52% and following a relatively normal distribution. Sharp GC content peaks and abnormal distribution in mutant samples 2 and 3 were observed with averages of 42-46% GC content (Figure 1C). High levels of sequence duplication were also observed with levels of duplication >100 reaching up to 16.58% across all samples (Figure 1D) . Despite concerns over potential bias in the library, the data was considered of a high enough quality to continue downstream analysis.

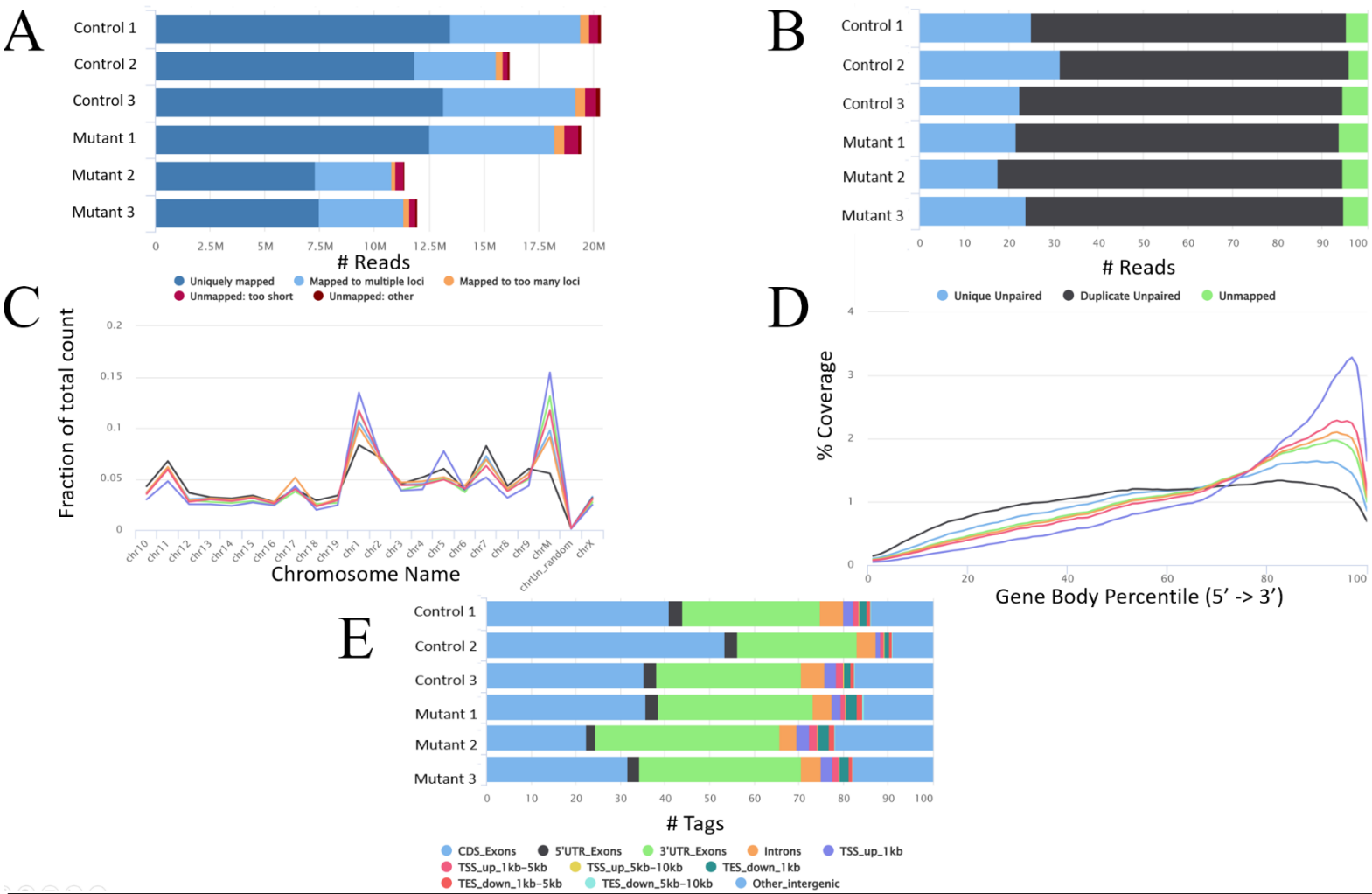

Figure 2. **Assessing Mapping Quality of RNA-Sequencing Reads to the mm9 Genome.** Summary bar and line plots of the RNA Star mapping quality analysis pipeline. STAR alignment scores across individual conditions are shown indicating the number of reads uniquely mapped in dark blue, multiple loci mapping in light blue, and rejected reads in orange and red (A). Reads categorised by duplication state shown as a % of each condition, with light blue representing uniquely mapped reads and black representing duplicated reads (B). Number of reads mapped per chromosome as a fraction of total count (C). Coverage of reads across the gene body displayed as a % from 5’ to 3’ for all conditions (D). Distribution of mapped reads over genomic features for individual conditions, with blue representing exonic mapping to coding regions , and black and green representing mapping to 5’ and 3’ untranslated regions respectively (E).

After mapping of the reads to the mm9 reference genome, the quality of mapping was assessed for all samples to determine the number of correctly mapped reads, levels of duplication after mapping, alignment of reads within the gene body, and distribution of reads across chromosomes and genomic features.

STAR alignment scores displayed an acceptable 62.4-73.3% level of uniquely mapped genes, but with a significant proportion of multi-mapping across samples, with 22.7-32.4% of reads being mapped to multiple loci (Figure 2A). Mark Duplicates validated the high level of duplication identified prior to mapping, as between 64.4-77% of reads were duplicated across all samples (Figure 2B). Analysis with Samtools idxstats identified an expected distribution of genes across most chromosomes. However, mitochondrial RNA was present at a rate higher than expected in both mutant and control conditions, reaching 15% of the dataset in mutant samples 2 and 3 (Figure 2C). Gene body coverage assessment of read locations showed a 3' bias across all samples, with the most exaggerated bias being observed in mutant sample 2 and 3 (Figure 2D). Read Distribution analysis of the mapped reads validated that 65-80% were mapped to exons, a typical number for RNA-seq investigations (Conesa et al., 2016). However, a significant number of these mapped reads were to 3' untranslated exons at a rate of 34.7-41.4% across all mutant samples (Figure 2E). This further validated the previously observed 3' bias displayed in Figure 2D.

Altogether, mapping was successful despite lower-than-expected levels of uniquely mapped reads and an observed 3' bias. Assessment of the mapping issues determined that they were not sufficient to stop progress to differential gene analysis.

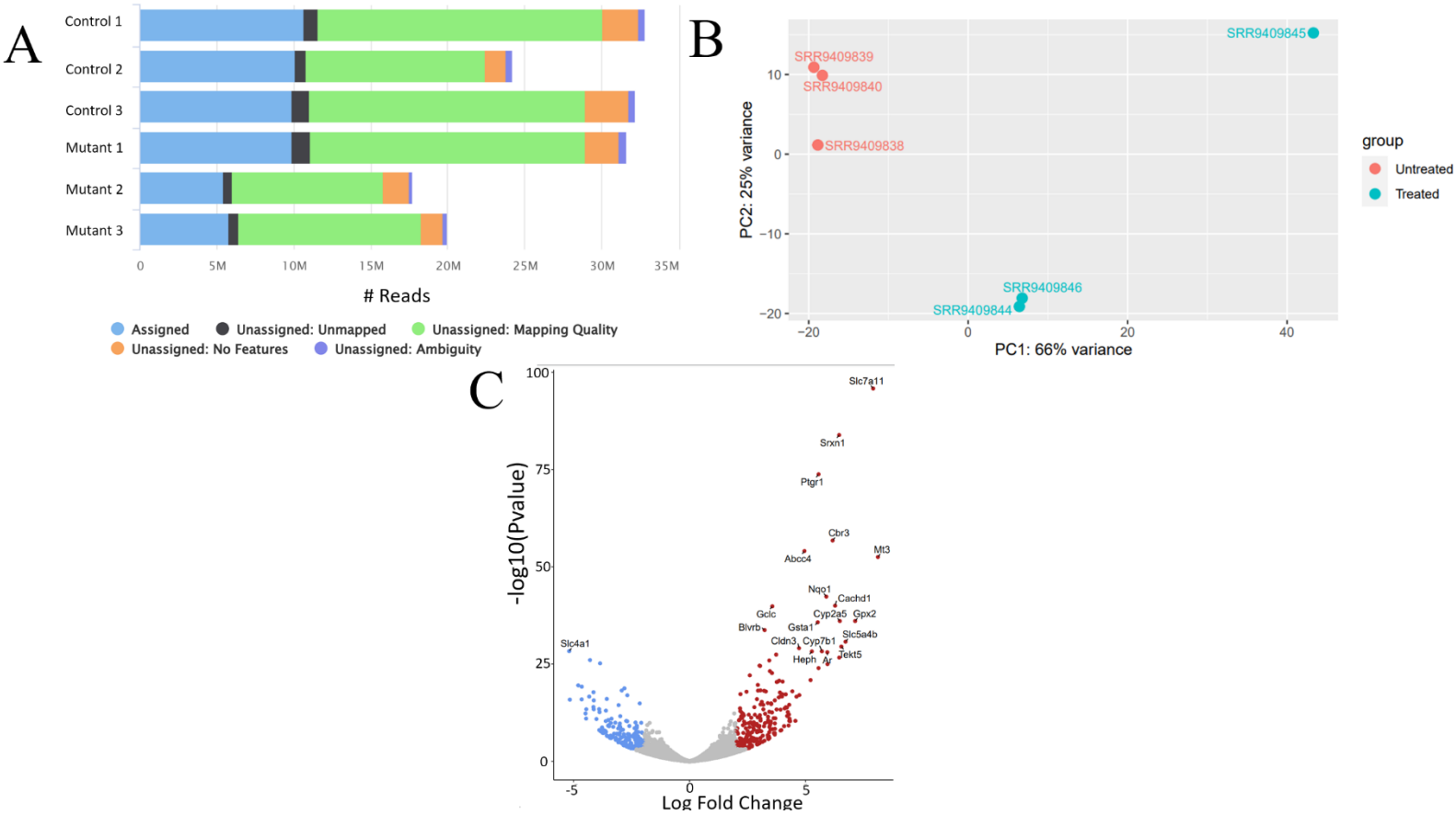

Figure 3. **Differential gene analysis using DESeq2.** featureCounts assessment of reads assigned to genomic features displayed for all control and mutant conditions, with the number of assigned reads displayed in blue, and unassigned or no feature reads being displayed in green and orange (A). Scatter principal component analysis (PCA) plot of transformed count data from DESeq2 with clustering according to treated or untreated conditions. Control datasets displayed as red dots and mutant datasets as blue (B). Volcano plot displaying -log10 P-Value of identified differentially expressed genes plotted against their log fold change. Significantly upregulated genes shown as red dots, significantly downregulated as blue dots, and genes of no significance as grey. Significance defined as $p < 0.01$ and >2 log fold change (C).

Following validation of mapping, mapped features were enumerated using featureCounts. Analysis of reads mapped to genomic features displayed a marked level of reads unassigned due to mapping quality with 29-41.9% of reads assigned to features across all samples, which reinforces previously displayed mapping issues (Figure 3A).

Principal component analysis (PCA) exhibited clustering and low variation within the control samples and mutant samples 1 and 3, with mutant sample 2 displaying significantly more variance in both principal component 1 and 2 (Figure 3B)

Changes in gene expression between the control and mutant condition were assessed using the DESeq2 software package to carry out differential analysis. 410 genes were found to be significantly differentially regulated within cut offs of <0.01 P Value and >2 log fold change. This was comprised of 255 upregulated genes and 155 downregulated genes, with expected ARE associated genes NQO1, GCLC, GSTP, and GPX identified as being upregulated amongst others (Ma, 2013; Figure 3C).

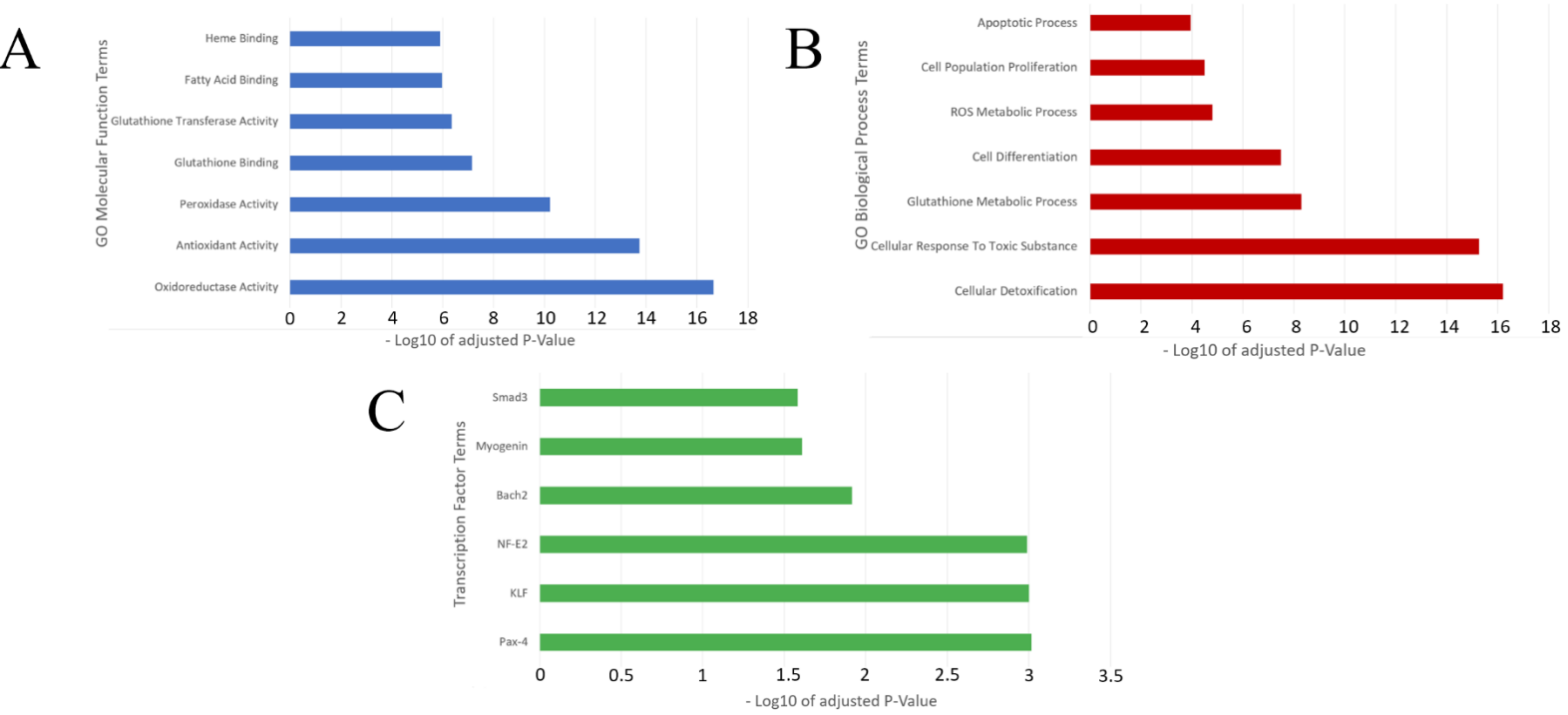

Figure 4. **Gene Ontology enrichment analysis of the differentially expressed genes via GProfiler.** Enriched gene ontology terms plotted against their -Log10 adjusted P-Value grouped by molecular function (A), biological processes (B) and enriched transcription factor binding sites (C). Molecular function and biological process lists are not exhaustive, with lowest P-adj value relevant terms displayed.

To link the identified differentially expressed genes with biological mechanisms, Gene ontology (GO) enrichment analysis was carried out using Gprofiler to derive their potential functions. GO terms were categorised according to their function and analysed using an adjusted P Value cut-off <0.01.

GO molecular function analysis revealed a strong enrichment for genes associated with responses to oxidative stress and antioxidant activity. Enriched terms included oxidoreductase activity, antioxidant activity, peroxidase activity, glutathione binding and transferase activity, antioxidant activity, and fatty acid binding (Figure 4A). Similarly, significant biological process GO terms found included many processes associated with response to cellular stress, such

as cellular detoxification and response to toxic substances. Further terms identified included a strong association with excess cell population proliferation, cell differentiation, and apoptotic process.(Figure 4B). Transcription factor binding site analysis found expected NF-E2 enrichment alongside Pax-4, KLF, Bach2, Myogenin, and Smad3 (Figure 4C).

Discussion

This project aimed to investigate the 'dark side' of NRF2 activation and provide insight into the safety of NRF2 as a potential therapeutic to treat chronic diseases. Overall, the findings of the project provide support for the recently discussed 'dark side' associated with NRF2 activation, as evidenced by the significant number of differentially expressed genes and GO terms that align with those previously discussed in the literature.

Initial data quality control with FastQC showed that the reads were of high quality (Figure 1A), but identified sharp GC content peaks (Figure 1C) and a high level of duplication (Figure 1D) that was validated after mapping to be between 64.4-77% across samples, as opposed to the expected 50% in RNA-sequencing data (UC Davis Genome Center, 2013; Figure 2B) . These quality issues may have been caused by ribosomal RNA contamination of the samples prior to generation of the RNA-sequencing reads (Yang et al., 2019). Alternatively, they could also be explained by a small number of highly expressed genes, which is common in RNA-sequencing data (Bansal, 2017). The signal from these genes, potentially those involved in antioxidant response, would dominate the signal from significantly differentially expressed genes that only showed a relatively small change in expression (Kukurba & Montgomery, 2015) This phenomenon, combined with the stringent Padj-value and log fold change significance thresholds that were necessary due to project time constraints, suggests that genuine true positives may have been excluded from this projects analysis. To support this conclusion, heme oxygenase 1 (HMOX1), a gene regularly used to validate NRF2 pathway activation, showed no significant differential expression at a 0.26 log fold change (Loboda et al., 2016).

Following mapping to the mm9 genome, mapping quality control identified a high level of mapping to multiple loci (Figure 2A), high mitochondrial RNA content (Figure 2C), and a distinct 3' bias (Figure 2D). A 3' bias in the data suggests that possible RNA degradation may have occurred during sequencing and library preparation, and that the datasets may be distorted towards shorter reads, especially in mutant samples 2 and 3 that displayed the most exaggerated 3' bias (Gallego Romero et al., 2014; Figure 2D). Shorter read length and lower depth of coverage in the library may explain the multi-mapping found by RNA Star as shorter reads are harder to determine mapping for (Li et al., 2014; Figure 2A).

Despite issues with duplication and 3' bias causing low levels of assignment to known genomic features, as identified by featureCounts (Figure 3A), there was a high level of mapping to exons within the genes that were assigned (Figure 2E). This combined with the high read quality assessed by FastQC indicated that the remaining data was adequate for differential gene expression and that meaningful conclusions could be drawn from further analysis. Analysis of RNA-sequencing data in future work could employ techniques such as the "RAUR" method to increase alignment of unmapped reads (Peng et al., 2015) and mmquant to include multi-mapped reads in quantification steps to ensure more comprehensive investigation of the dataset (Zytnicki, 2017).

Of the 410 differentially expressed genes identified by DESeq2 analysis and pathways identified by gene ontology enrichment analysis, the majority were known and expected NRF2 pathways such as antioxidant activity, oxidoreductase activity, cellular detoxification, and ROS metabolism (Ahmed et al., 2017). This provides support for the efficacy of proposed therapeutics that inhibit Keap1 to activate NRF2 as the antioxidant effects anticipated in these studies have been validated (Lu et al., 2019). However, other significant processes and pathways were identified that provided an insight into the 'dark side' of NRF2 recently discussed in the literature (Wang et al., 2008; Rojo de la Vega et al., 2018; Orrù et al., 2020; Zang et al., 2020).

One of the processes identified by GO enrichment was the upregulation of glutathione transferase and binding activity (Figure 4A). Genes associated with these terms such as GTSP1/2 and GTSA1 were identified to be upregulated with a log fold change of 4.1-5.53. Recent studies have shown modulation of these genes to be promising predictors of hepatic dysfunction, and have linked them to the development of neoplastic disease (Orrù et al., 2020; Prysyazhnyuk, 2021). This validates previously discussed studies that suggest NRF2 could be an oncogene (Menegon et al., 2016). These results are particularly significant due to the conditions of this project involving reads from hepatocytes, suggesting a valid 'dark side' of NRF2 activation has been identified in the theorised target tissue, with implications for NRF2 therapeutic development and potential side effects.

Enrichment for GO terms for cell proliferation and differentiation were also identified by analysis (Figure 4B). Genes associated with these terms such as BMPR1a, a known NRF2 target gene related to cell cycle and proliferation (Wu et al., 2019), and CASP3, a gene involved in control of apoptosis (Porter & Jänicke, 1999), were upregulated and downregulated respectively. These observed effects alongside enrichment for upregulated cellular detoxification further support the proposed 'dark side' of NRF2 activation in respect to it conferring chemoresistance and growth advantage in cancers (Wang et al., 2008).

During GProfiler analysis of enrichment for significant transcription factor binding sites, many were observed including the expected NF-E2, Pax4, KLF and Bach2 (Figure 4C). This suggests that not all observed gene expression changes were directly due to NRF2 transcription, but by activation of a second level of factors. Future work is needed to identify the relationship between these secondary factors and possible off target side effects that could occur due to NRF2 therapy, and may provide insight into why trials of drugs such as bardoxolone exhibited negative reactions that caused trial termination (Tayek & Kalantar-Zadeh, 2013).

Overall, it is clear from this projects analysis that the 'dark side' of NRF2 has merit and warrants further investigation. Many diseases potentially targeted by NRF2 therapeutics such as multiple sclerosis (Carlström et al., 2019) and Parkinson's (Yang et al., 2022) are positively correlated with age (Wyss-Coray, 2016), as is the development of cancer (White, 2014). Additionally, liver dysfunction is a common comorbidity in many of the targeted conditions (Peng et al., 2022). Therefore, careful patient selection and future investigations would be recommended to ensure the identified 'dark side' of NRF2 that drives cancer progression and hepatic dysfunction does not negatively impact patients who may be at elevated risk of either disorder. Future studies should compare this project's results with conditions that more closely resemble realistic NRF2 overexpression, perhaps through the use of proposed therapeutics such as Bach1 inhibitors (Ahuja et al., 2021), and include histological components where the livers of treatment condition mice are examined for morphological changes induced by the identified cell proliferation genes.

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